

Ayush Singh

1511.ayushsingh@gmail.com | <https://github.com/AyushSingh331> |
<https://www.linkedin.com/in/ayush-singh-36a71a299/> |
<https://leetcode.com/u/Endeavour15/>

EDUCATION

Vellore Institute Of Technology

Bachelor of Technology in Computer Science (Specialization in AI & ML)

August 2023 – May 2027

GPA: 8.66

TECHNICAL SKILLS

Languages: Java, SQL

Data Science: NumPy, Pandas, Matplotlib, Scikit-learn

PROJECTS

EchoText

<https://github.com/AyushSingh331/Speech-To-Text-and-Speech-Translator>

- **Fine-Tuned ASR Model:** Fine-tuned a pretrained HuBERT-Large speech recognition model on the LJ Speech corpus using CTC loss, reducing Word Error Rate to 6.7% over 9,500 training steps.
- **Cascaded Inference Pipeline:** Architected a two-stage inference pipeline combining a spoken language identification model with the fine-tuned ASR model, gating transcription to confidently-detected English audio and preventing invalid output on unsupported languages.

Breast Cancer Prediction & Prevention Platform

<https://github.com/AyushSingh331/Breast-Cancer-Prediction-and-Prevention>

- **Deep Learning Transfer Learning:** Implemented and fine-tuned an EfficientNetB3 Convolutional Neural Network (CNN) architecture in TensorFlow to perform high-accuracy medical image classification.
- **Model Evaluation Optimization:** Applied hyperparameter tuning, learning rate scheduling, and data augmentation techniques to maximize precision and recall metrics on medical datasets.

CERTIFICATIONS

Hackathons

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- Ranked in the Top 500 at the Amazon ML Challenge and secured a round two qualification in the Adobe Hackathon.

MP Online AI Foundation Certificate

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- Completed comprehensive coursework covering foundational concepts in Artificial Intelligence, Machine Learning and their real-world applications.